Jin-gyun Kim, Ph.D.,
Associate Professor,
Department of Mechanical Engineering,
Kyung Hee University
1732 Deogyeong-daero,
Giheung-gu, Yongin-si, 17104, Gyeonggi-do
South Korea
E-mail: jingyun.kim@khu.ac.kr
Web: sites.google.com/site/modelingnsimulation

3 November, 2025

Letter of Recommendation of *Hyeonbeen (Edward) Lee*

Dear Admission Committee,

I am pleased to give my strong recommendation for Hyeonbeen (Edward) Lee for admission to the Computer Science Ph.D. program at Stanford University. With five years of close supervision and mentorship since 2021, I witnessed his entire academic journey—from a student in my class to an undergraduate researcher, a Master's student, and a postgraduate researcher. Through this prolonged interaction, I have developed solid confidence that Lee possesses outstanding intellectual curiosity, technical skills, and resilience required to excel in your program and as an independent researcher.

In my courses on computational dynamics and finite element methods, he was one of the finest students and consistently achieved top grades by excelling at understanding complex formulations and their computational implementations. He was especially impressive for his proactive analyses beyond the required tasks, such as using symbolic math libraries to demonstrate his derivation, or translating MATLAB codes into Python and sharing them via GitHub for reproducibility, all of which went well beyond the course material. These self-motivated activities, driven by his intellectual curiosity, strongly validated his potential as a researcher, which was my key reason to accept him as an undergraduate researcher.

After joining the lab, he showed an intense interest in *machine learning* and *neural networks*. He was particularly passionate and proficient in our research on data-driven simulation. As an undergraduate researcher, he rapidly digested relevant publications and reproduced key results within months, while also leading a machine learning thesis project with his undergraduate co-advisor Prof. Shinkyu Jeong.

He always sought to produce something beyond the given task, which distinguished him from other students. While co-working with our team on the vehicle dynamics simulation and neural network-based meta-modeling, he identified the limitation of our existing approach in modeling impulsive dynamics and proposed a novel multi-order learning architecture. This work eventually led to his first-author publication in *Journal of Computational Physics*, which also received the Excellent Paper Award from the Korean Society of Mechanical Engineers conference.

At the end of his first Master's year, he also started working on the project of ML-based joint force estimation for a marine excavation robot. Our initial objective was a single-step force prediction in a simulated environment. However, within a year of him taking on the project, Lee progressively

expanded the project by shifting to real-world sensor data, applying sequential modeling, and reframing it as a forecasting problem. Consequently, he came up with a decomposition-based generative forecasting framework for contact-rich real-world force dynamics. These improvements yielded not only an impressive advance in the model's accuracy but also served as a valuable foundation for the lab's machine learning capability and led to subsequent success in the project.

After earning his Master's degree, Lee continued on this project as a postgraduate researcher. For two years, he dedicated his effort to accelerate the model's inference for real-time application through systematic optimization, along with in-depth refinement of the model by testing and integrating state-of-the-art learning methodologies. Concurrently, he also enthusiastically mentored a junior master's student who was building upon his work.

Besides his research excellence, he also maintained good relationships with the other lab members and shared his knowledge with the team. He created the lab's GitHub, led systematic code management, and actively shared well-documented repositories to support his peers. Furthermore, he used to get along well with international and visiting lab members and helped them overcome environmental hardships, all of which can demonstrate his emotional maturity and team spirit.

In summary, Lee possesses an outstanding combination of intellectual passion, validated research skills, and collaborative spirit. He is one of the most impressive students I have mentored, and I am confident he will make an excellent contribution to your Ph.D. program. Therefore, I recommend him without any reservation.

Sincerely,

Jin-gyun Kim,
Associate Professor,
Department of Mechanical Engineering,
Kyung Hee University